

Vibe Theory

The Experimental Model

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June 6, 2026

Abstract

This paper gives a technical account of Vibe Theory as it now stands: the mathematical model, and a walk through thirty-six findings (P1 to P36) from a finite, reproducible testbed. The model is monist: it treats reality as a single kind of entity, the *vibe*, a unit of experience carrying a ternary *tone* and related to others only by the *note*, on a growing, random, hyperbolic, causal mesh, updated by a local ternary signed-majority rule. There are no continuous weights and no hidden state, and every continuous quantity of physics is an emergent, coarse-grained description of the discrete substrate. The findings span the framework: geometry recovered sharply and made the dominant phase of a measured transition, the law as a trilemma resolved by the emergent-mesh Hamiltonian, a Lorentz-safe substrate, the four field spins with mass from a Higgs and the gauge and graviton operators derived from the action, the Newtonian limit and a conserving Einstein equation, the quantum pillars, and a cosmology with an arrow, expansion, and a dark sector. Three predictions make contact with data, most notably the everpresent cosmological constant matching the observed dark energy to order of magnitude. Finally the model is run end-to-end, with the key emergent structures coming off one instantiation. We are explicit about what is demonstrated, what is a first rung on a deep problem, and what is open. This is a simulable model with a concrete build target, not a finished empirical theory.

Status and caveats. This is exploratory work, a feasibility study rather than a validated result. The aim is to see whether a model of this kind can be made to hang together at all, and so far it looks promising. Much validation and further work remain. Several of the experiments reported here are preliminary, and some may turn out to be inaccurate or improperly set up. Nothing here should be read as established or taken too rigorously at this stage. It is offered in the spirit of curiosity and invitation, a direction that looks worth exploring rather than a finished theory or a claim of discovery.

1. Introduction

This is the fourth presentation of Vibe Theory and the most complete. The first (Pollard, 2025) set out the framework as philosophy. The second (Pollard, 2026a) gave it a discrete mathematical structure. The third (Pollard, 2026b) tested the conceptual core in simulation. This paper does two things at once. It states the model in its current form, and it walks through the full set of findings, thirty-six in all, saying for each what was found and where we landed.

The thesis is monism. The framework treats reality as a single kind of thing, the vibe, a unit of experience, whose only act is to experience other vibes and be experienced by them. In the model there is no background space or time. Space is then the shape of the web of vibes seen from far away, time is the depth of its before-and-after, and everything else, matter, force, the quantum, is modeled as a pattern in the web. A monist theory of this strictness owes a debt most ontologies never pay: it must show that the variety of the physical world actually comes out of one substrate, not merely assert that it could. This paper pays that debt as far as a finite simulation can.

The method is to turn each load-bearing question into a runnable measurement. We built a testbed, a finite and seeded program with a known-answer test suite, and posed the questions the framework must answer to be coherent, each as an experiment that returns a number and that could fail. Several early versions did fail, in plausible-looking ways, and were caught by the known-answer tests and corrected. What survives is reported here, sorted honestly into what is demonstrated, what is a first rung on a deep problem, and what is open or beyond reach.

The paper is organised as follows. Section 3 states the current model, the substance, the relation, the rule, and the growth, with the small constructor that writes it at a glance. Section 4 describes the testbed and the known-answer discipline that makes its numbers trustworthy. The result sections then walk through the findings by theme: foundations (geometry, time, the law, the dominant spacetime phase), Lorentz invariance, matter and the fields, forces and gravity, the quantum sector, cosmology and the dark sector, holography, the distinctive predictions and their contact with data, and the capstone where the whole model is run as one system. Section 14 is the full scoreboard of all thirty-six findings with their honest status.

Throughout, the framework sits in the family of discrete-physics programs, the causal sets of Bombelli, Lee, Meyer, and Sorkin (1987), the lattice gauge theory of Wilson (1974), the overlap fermions of Neuberger (1998), and the cellular-automaton interpretation of quantum mechanics of 't Hooft (2016). What Vibe Theory adds is the insistence that all of these live on one experiential substrate, and the demonstration, here, that they can. This work is a philosophical and mathematical model with a simulable build target rather than a finished scientific theory. References to physics are sources of structure and tests of coherence, not claims to have derived nature.

2. The Shape of the Model: A Random Hyperbolic Network

Before the formal definitions, the picture in plain terms, since the whole framework rests on one unfamiliar kind of object, and the natural guesses about it are all wrong in instructive ways.

What it is. The mathematical structure is a graph that grows. Picture a vast web of points, each point a vibe, joined by links, each link a note. The web is not a grid. Its points are scattered at random, and the space they are scattered in is hyperbolic, negatively curved, the saddle-shaped geometry in which the room around any point keeps opening up the further you look, so the number of points within reach grows exponentially with distance. On this web sits a simple rule that resets the state of each point from its neighbours, run over and over, while the web itself keeps adding points at its leading edge. In the language of discrete physics this is a causal set carrying a network automaton: a random, hyperbolic, growing graph with a local update rule. That object, and nothing richer, is the model.

How it works. Each point carries a single three-valued state, its tone, one of $\{-1, 0, +1\}$, read as pain, peace, and pleasure. It is joined to about ten others, not a fixed number but a spread, the ones it directly experiences. Each link carries its own three-valued number, the fill, the strength and sign of that attention. To update, a point reads the outward states (the wills) of its neighbours, multiplies each by the fill of the link, adds the results into a single whole number, and takes the sign of that sum as its next state: a positive total becomes pleasure, a negative one pain, a tie peace. There are no fractional weights and no stored real numbers anywhere, only three-valued states and a transient integer count, exactly the spirit of a cellular automaton's tally of its neighbours. The points update one at a time with no global clock, and the web only ever grows, its links accumulating and never undone.

What it is not. It is not a regular cellular automaton on a square or cubic grid. A grid has preferred directions, so on a grid the speed of light would depend on direction and Lorentz invariance would break, which experiment forbids. It is not a regular hyperbolic tiling either, the orderly Escher-like packings of the hyperbolic plane, for the same reason, they are still too rigid. The randomness is not a convenience but the load-bearing choice, since only a random scatter has no preferred frame even at the smallest scale. It is not a field theory on a smooth space, because there is no smooth space underneath. And there are no hidden variables behind the tones, and no continuous amplitudes stored on any point.

Where the smooth physics lives. If nothing continuous is stored, where do the smooth numbers of physics come from? They are large-scale summaries of many points, the way a temperature summarises jiggling molecules. The geometry of spacetime is the shape of the web seen from far off. Energy and the flow of time are what a local operator, the graph's Laplacian, does on the grown web. A quantum amplitude is a patch of many points read together, its magnitude the fraction of them excited, its phase their shared rhythm. Each of these is real and measurable, and none is written on any single vibe. The base is discrete and three-valued, the continuum is the view from a distance.

This is the object the next section defines precisely and the rest of the paper tests: a growing, random, hyperbolic network of three-valued points that vote locally, with all the smooth structure of the world appearing only in the large.

3. The Current Model

We state the present model precisely. It has a substance, a relation, a state, a rule, and a growth. Nothing else.

3.1. The substance and the relation

Definition 1 (Vibe and tone) A *vibe* is the one kind of entity, a unit of experience. Each vibe v carries a *tone* t_v , its felt quality, drawn from the ternary alphabet

$$t_v \in \{-1, 0, +1\},$$

read as pain, peace, and pleasure. The tone is not an object inside the vibe. It is the vibe seen as a felt state.

Definition 2 (Note, will, and fill) The *note* is the only relation: v noting w is v experiencing w . A note is not a third object between two vibes, it is one vibe's act of experiencing another, and read from the outside it is an edge of a graph. Each vibe v has a neighbourhood $N(v)$ of the vibes it notes. A note carries two ternary quantities. The *will* of w , $\omega(t_w) \in \{-1, 0, +1\}$, is its tone turned outward, what v reads when it notes w . The *fill* $f_{vw} \in \{-1, 0, +1\}$ is the strength and sign of the attention along the note, itself the tone of the relational vibe carrying that note, and symmetric, $f_{vw} = f_{wv}$, since a note is one shared relational vibe.

There are no continuous weights anywhere, and no hidden state. Everything stored is a ternary tone, on vibes and on the relational vibes that carry the fills.

3.2. The substrate

Definition 3 (The mesh) The vibes and their notes form the *mesh*, a graph. The chosen mesh is a *random, hyperbolic, causal* graph: nodes are placed by the hyperbolic area measure and joined by hyperbolic proximity, so the degree is variable (mean about ten), not a fixed lattice coordination. The randomness makes the mesh Lorentz-invariant in distribution, with no preferred frame even at the discreteness scale. The hyperbolicity makes it at once exponentially reaching, Lorentz-safe, and navigable without a global address. The order of the notes is a locally finite partial order, a causal set.

This is not a regular cellular automaton (a lattice has a preferred frame), nor a regular hyperbolic tiling (still too regular). It is the random hyperbolic causal mesh, the choice forced by Lorentz invariance (Section 6).

3.3. The rule

Definition 4 (The signed-majority update) Time advances in discrete *beats*. Each beat, asynchronously and locally, a vibe v sets its next tone to

$$t_v \leftarrow \text{sign}(h_v), \quad h_v = \sum_{w \in N(v)} f_{vw} \omega(t_w),$$

where sign returns $+1$, -1 , or 0 on a positive, negative, or zero field. The local field h_v is a transient integer in $[-\deg v, +\deg v]$, computed during the beat and discarded, so it is not hidden continuous state, exactly as a cellular automaton's neighbour count is transient. The contribution of each neighbour is the ternary product $f_{vw} \omega(t_w) \in \{-1, 0, +1\}$: a fill of $+1$ takes the neighbour's will as it is, -1 inverts it, 0 ignores it.

This is the ternary analogue of an Ising or Hopfield update: ternary states, ternary couplings, the sign of the local field. With symmetric fills and asynchronous updating it flows to stable structured states (Section 13), the basis for persistent matter and selves.

3.4. The growth

Definition 5 (Eternal growth) The mesh grows forever. New vibes are born to the future of a small local neighbourhood of the leading edge, inheriting its past. With a net birth above one per region the universe expands. The accumulation is irreversible, distinctions only pile up, and that pile-up is the arrow of time.

3.5. Emergent continuous quantities

A single vibe is strictly ternary and carries no continuous quantity. Every continuous object of physics is emergent, a coarse-grained description over many vibes, never a number stored on a vibe.

Principle 1 (Discrete base, emergent continuum) The geometry, the Hamiltonian and its eigenvalues, and the quantum amplitude are all large-scale effective descriptions of the ternary substrate. A quantum amplitude $\psi(x)$ at a coarse-grained point x (a patch of many vibes) has magnitude $|\psi(x)|^2$ equal to a density, the fraction of vibes in the patch that are coherently excited, continuous only in the large-patch limit, and phase equal to the patch’s collective clock-tone. This is the same status a continuum wavefunction has over any discrete substrate, or a magnetisation has over discrete spins.

3.6. The model at a glance

The whole model is written in a few lines by a small constructor (the experiment of Section 13), so the description and the implementation are one object. With no options it is the present model, and one option changed expresses a variant for comparison:

```
vibe model
  mesh      hyperbolic      random hyperbolic causal mesh, mean degree ~10
  tone      ternary         {-1, 0, +1}, the felt quality of a vibe
  fill      ternary-symmetric each note carries a shared ternary fill
  rule      signed-majority  next(v) = sign( sum_w fill(v,w) * will(w) )
  schedule  asynchronous    local, neighbours only, no global clock
  growth    net-positive     eternal expansion by local birth
```

3.7. Committed, working, and open

For honesty, every choice carries a status, used throughout the paper.

- **Held fixed for now:** monism, discreteness, the note as the only relation, ternary tone, the random hyperbolic mesh, local asynchronous updates, the signed-majority (sign-of-integer-field) rule with no continuous weights, variable connectivity, eternal growth, no hidden state.
- **Working (current best, alternatives open):** the mean degree (about ten), the exact tie-breaking and schedule, the emergent clock-tone phase, the birth probabilities, the smearing scale of the action.
- **Open (not decided):** the derivation of the Born rule, the full nonlinear Einstein equation, the Standard-Model content, and the framework’s foundational claim about experience.

4. The Testbed and Why Its Numbers Are Trustworthy

The model is fully discrete, so it is fully simulable. The testbed is an open-source TypeScript library (the `@cluesurf/vibe` package). It generates a substrate, runs a local rule over it in discrete beats, and measures the emergent physics. Everything is finite and seeded, so every number in this paper is a deterministic function of a stated seed and reproduces exactly. Real numbers appear only as measured outputs (coordinates, eigenvalues), never as the base.

At the core is a suite of *known-answer tests*. A number on its own means nothing. A number from a tool checked against a case with an independently known answer does. Every measurement tool is calibrated against such a case: a Poisson sprinkling of a d -dimensional region must measure dimension d , a Bell test with independent settings must respect the classical bound of 2, the overlap operator must have exactly one fermion species with machine-zero chiral-symmetry residual, the uniform-measure sampler must reproduce exact enumeration, and so on. There are fifty-six such checks, and they all pass. Several early versions were wrong in plausible-looking ways, a miscalibrated dimension estimator, a sprinkler not uniform by proper volume, a Monte Carlo move that did not sample the right measure, and the known-answer tests caught them. A program that cannot be caught being wrong is not science. This one can be, and was, repeatedly.

What the testbed establishes falls in three honest categories. First, the machinery is correct, validated by the known-answer tests. Second, the framework can carry known physics: reproducing the Newtonian potential, the photon, the graviton, confinement, and the electroweak masses shows the mesh is expressive enough to hold the real world, which is a load-bearing test, since a monist theory is falsified if matter, force, and gravity cannot be built from one substrate. It is not, by itself, proof the framework is uniquely correct, since standard physics reproduces these too. Third, the genuinely distinctive claims, where the framework makes a stand of its own: the both-worlds substrate, the dynamical selection of spacetime, the superdeterministic quantum tension, the Lorentz-safety of random discreteness, and the everpresent cosmological constant. The distinctive predictions that meet data (Section 12) are what could confirm or falsify the theory against nature.

5. Foundations: Geometry, Time, the Law, the Dominant Phase

5.1. The geometry is unique (P5)

A causal set fixes its continuum geometry through the *Hauptvermutung*: order plus number recovers a Lorentzian manifold essentially uniquely. Eight independent sprinklings of the same region recover the dimension as 3.02 ± 0.05 , and the proper-time distance across the region has coefficient of variation 0.027. **Where we landed:** the smooth geometry a discrete order carries is sharp, not just in dimension but in distance. The conjecture itself remains unproven, here and in the literature.

5.2. Spacetime is the dominant phase (P2, P6, P12)

Most causal sets are not manifold-like, the layered Kleitman-Rothschild orders dominate the uniform measure by sheer number. A theory in which spacetime is real must show its dynamics defeats them. The sharp Benincasa-Dowker action fails (it drives the ensemble to layered orders). The *smeared* action fixes it. The decisive tool was a correct uniform-measure sampler (single-pair toggle, accept only if still transitive), validated exactly against enumeration, which reaches the large- N entropic regime where the layered orders take over by $N = 64$. There the smeared action makes manifold-like spacetime a stable phase coexisting with the layered one, a first-order signature holding across $N = 48$ to 160, and the 2D specialisation (P6) is genuinely two-dimensional (Myrheim-Meyer dimension 2.0 to 2.1). Finally a Wang-Landau estimate of the density of states (the $1/t$ schedule) measures the free-energy crossing (P12): $\beta^* \approx 0.14$, roughly N -independent. **Where we landed:** the dynamics does not merely permit smooth spacetime, it makes it the *dominant* phase of the sum over histories above a finite coupling, with the layered phase metastable. The remaining nuance is

the free-energy crossing at large N , which needs a height-changing cluster move (the single-pair move cannot extend a chain).

5.3. The law is a trilemma, resolved by the emergent Hamiltonian (P1)

Is the dynamics governed by a Hamiltonian that is both bounded below and local? For a reversible rule's own logarithm $H = i \log U$ the answer is a trilemma: local, bounded-below, and information-propagating cannot all three hold, any two can. The principal Hamiltonian of a nontrivial reversible rule is nonlocal (range about $0.8N$), and the explicit local branch exists only for non-propagating commuting-gate rules. **Where we landed:** the resolution is to stop using the log of a microscopic step. The emergent-mesh Hamiltonian, the graph Laplacian or the Dirac operator on the grown mesh, is local (range 1, size-independent), bounded below (spectrum from 0), and propagating (a finite-speed lightcone) all at once. The rule builds the geometry, and time is what the local operator does on it. The rule and emergent time are distinct objects.

6. Lorentz Invariance (P3, P11, P27)

6.1. The both-worlds substrate (P3)

The substrate must be two things that pull against each other: Lorentz-safe (no preferred frame, which random structures give but rigid ones do not) and navigable (routable from local information, which addressable rigid structures give but generic random ones do not). **Where we landed:** the connected hyperbolic random graph is both at once. With mean degree about eleven it has exponential reach, anisotropy 0.07 (as Lorentz-safe as a pure sprinkling), and 100 percent backtracking navigability, and all three survive an eightfold growth of the mesh at constant density. The price is greedy geometric routing instead of exact addressing, consistent with no hidden state. This is the result that fixes the chosen substrate.

6.2. The dynamics is Lorentz-isotropic in the infrared (P11)

Does the emergent dynamics propagate isotropically? **Where we landed, with an honest correction:** both a lattice and a random mesh propagate isotropically at long wavelength. This is the lattice-field-theory fact that rotational and Lorentz invariance emerge in the infrared even on a lattice, the anisotropy being a lattice-scale effect. So the clean substrate-level distinction is P3, not the low-energy dynamics, and discreteness threatens Lorentz invariance only at the Planck scale.

6.3. Random discreteness is Lorentz-safe, a lattice is not (P27)

At the discreteness scale the difference is decisive. A lattice has energy-dependent, directional Lorentz violation: its group-speed anisotropy grows with energy (from 0.003 to 0.71 across the band) and the speed of light falls below one. A random sprinkling has none: its nearest-neighbour link directions are isotropic (a 4-fold anisotropy of 0.04 versus 1.0 for the lattice). **Where we landed:** because the framework uses the random sprinkling, it predicts *no* first-order Lorentz violation, which is exactly what gamma-ray-burst photon timing observes (Section 12). Random discreteness preserves Lorentz invariance where regular discreteness breaks it, and the residual effect of discreteness is not coherent violation but the stochastic swerve of Section 12.

7. Matter and the Fields (P4, P14, P20, P22, P23, P25)

7.1. The monist spinor and spin from topology (P4)

Matter is built from the same tones, with no new primitive. A spinor is a graded complex of ternary vibes (vertices, edges as relational vibes, faces), not a richer tone on one vibe, with its complex phase emergent from a finite clock-tone. The Kahler-Dirac zero-mode count equals the surface Betti sum exactly: 1 for a disk, 2 for a cylinder, 4 for a torus. **Where we landed:** matter is monist and spin is topological. The chirality wall (Nielsen-Ninomiya) is threaded: the overlap operator gives a single fermion species with exact lattice chiral symmetry (Ginsparg-Wilson residual 6×10^{-16} , machine zero), where the naive operator has four doublers. This is at the effective (Dirac-on-a-complex) level, the bottom-up ternary-cluster construction being the open frontier.

7.2. Mass and the relativistic dispersion (P14)

A mass term in the lattice Dirac Hamiltonian opens a spectral gap equal to the rest energy and bends the dispersion into the relativistic form. **Where we landed:** the gap equals the mass exactly, the dispersion is $\omega^2 = k^2 + m^2$ (slope near one, intercept m^2), and at $m = 0$ the gap vanishes and $\omega = |k|$, a massless light-cone mode.

7.3. The photon (P20) and the gauge operator from the action (P23)

The free U(1) gauge field is a proper photon: on a periodic lattice about one third of its modes are exact gauge zero modes (the gauge freedom) and the rest are the two transverse polarizations, with a massless spectrum (the smallest physical ω^2 shrinks as the lattice grows) that a mass term would gap. **Where we landed:** the photon operator is not put in by hand. It is the small-field limit of the Wilson gauge action, $1 - \cos \theta \rightarrow \frac{1}{2}\theta^2$: the ratio of the Wilson action to the Maxwell action converges to one as the field shrinks (0.926, 0.988, 0.997, 0.9999). So the gauge kinetic operator is derived from the discrete action.

7.4. The Higgs (P22) and electroweak breaking (P25)

Mass is generated, not inserted. A scalar with a Mexican-hat potential settles at a nonzero vacuum value v (the symmetric phase has $v = 0$, the broken phase $v = 1$), and a gauge field coupled to it eats the Goldstone mode and becomes massive: the photon gap goes from 0 to exactly $(gv)^2$. **Where we landed:** this extends to the specific Standard-Model pattern. A Higgs doublet breaks $SU(2) \times U(1)_Y$ to $U(1)_{\text{EM}}$, giving the W and Z mass and leaving the photon massless, with the Weinberg relation $m_W = m_Z \cos \theta_W$. With the measured couplings it reproduces the observed masses: $m_W = 80.0$ GeV, $m_Z = 91.3$ GeV, $\sin^2 \theta_W = 0.233$. So scalar, spinor, and vector fields are all present, with mass from symmetry breaking. The rest of the Standard-Model content (three generations, the Yukawa spectrum, the gauge group itself) is the larger program.

8. Forces and Gravity (P8, P16, P21, P24, P28, P32, P33)

8.1. The gauge force (P8)

A force is the twist a tone gets crossing a note, a gauge connection on the links. **Where we landed:** 3D SU(2) lattice gauge theory confines (the string tension is positive at every coupling, falling from 1.32 to 0.40,

the Wilson area law). The chiral overlap fermion sees gauge topology exactly: in a $U(1)$ background of charge Q its index equals $-Q$, the lattice Atiyah-Singer theorem. And in a dynamical gauge field a chiral condensate forms from the anomaly, in both the Abelian Schwinger model and the non-Abelian $SU(2)$ case. The open rung is the Weyl-projected chiral gauge theory, unsolved in physics generally.

8.2. The Newtonian limit (P16)

The static potential is the Green's function of the emergent Laplacian. **Where we landed:** it matches the continuum in every dimension, confining in 1D, logarithmic in 2D, and Newtonian $1/r$ in 3D (the $1/r$ form is the best fit at $R^2 = 0.997$, beating $1/r^2$ and $\log$).

8.3. The graviton, and its operator from the action (P21, P24)

The graviton is the propagating excitation of the geometry, a symmetric rank-2 tensor. **Where we landed:** it is a proper massless spin-2 field with exactly two transverse-traceless polarizations (a massive spin-2 would have five), and a massless dispersion. And its kinetic operator is derived from the action: the linearized Einstein operator (the second variation of the Einstein-Hilbert action, the continuum limit of the discrete Benincasa-Dowker action) is diffeomorphism-invariant (pure-gauge perturbations are annihilated to machine zero) with exactly two massless modes at eigenvalue $\frac{1}{2}|k|^2$, no projector imposed.

8.4. The Einstein equation as an equation of motion (P32)

Where we landed: the linearized Einstein equation behaves as a real theory of gravity. The Einstein tensor is transverse, $k \cdot G[h] = 0$ to machine zero, so the equation automatically enforces energy-momentum conservation. It reduces to Newton in the static limit (the Poisson equation, P16), and in vacuum it propagates a massless graviton at the speed of light ($\omega/|k| = 1$ at every frequency). The full nonlinear Einstein equation from the discrete action is the remaining step.

8.5. Singularity resolution (P28)

Where we landed: discreteness gives a minimum causal length (it scales as $\rho^{-1/2}$ in 2D), so curvature, which goes as one over length squared, is capped at a finite value that rises with density and never diverges. The big-bang and black-hole singularities of general relativity are resolved: a finite-density beginning at the discreteness scale, not an infinite-density point, which fits the bootstrap (the universe starts from a first element, not a singular $t = 0$).

8.6. Black-hole entropy (P33)

Where we landed: the entanglement entropy of a region scales with its horizon *area*, not its volume (a 3D area fit beats a volume fit). Reading the region as a black hole and its boundary as the horizon, this is the Bekenstein-Hawking law $S = A/4$, with the entanglement across the horizon as its microscopic origin. The full black-hole physics (Hawking radiation, the exact coefficient, the information question) is the long road ahead.

9. The Quantum Sector (P7, P17, P31)

9.1. Quantum statistics from a classical base (P7)

Bell's theorem says no local model of independent coins matches the quantum correlations unless the measurement settings are not statistically independent of the hidden state. A monist mesh with no hidden state denies exactly that independence, since the settings are made of the same substrate as the system. **Where we landed, quantified:** as the setting-state correlation rises, the CHSH value climbs 1.0, 1.6, 2.3, 3.1, 4.0, crossing the classical bound and passing the quantum value. The currency is aligned bits, not bits: one bit of aligned dependence reaches $S = 4$, one bit misaligned gives $S = 1$. The honest residual: in a natural mesh the shared past of two spacelike measurements shrinks with separation, so the violation decays with distance ($4.0 \rightarrow 2.58 \rightarrow 1.04$), while quantum violation does not. The framework makes Bell violation possible from a classical base and pins the exact remaining puzzle (why the correlation is aligned and separation-independent), rather than hiding it.

9.2. Quantum coherence (P17)

Where we landed: a continuous-time quantum walk on the mesh spreads ballistically (width $\sim t^{1.00}$) while a classical walk diffuses (width $\sim t^{0.50}$). Genuine interference, the heart of quantum behaviour, emerges from the unitary evolution e^{-iHt} on the mesh.

9.3. The pillars of the formalism (P31)

Where we landed: the three pillars are present on the mesh. Unitarity, the total probability $\sum |\psi|^2$ is conserved at 1.000000. Interference, the probability of a sum of amplitudes differs from the sum of probabilities by a nonzero cross term, so amplitudes add, not probabilities. And a conserved Born probability, $|\psi|^2$ non-negative and summing to one. This is a down-payment, not the full theory. The open problem is to derive why the probability is $|\psi|^2$ and the measurement rule from the mesh itself, building on the superdeterministic hinge of P7.

10. Cosmology and the Dark Sector (P10, P13, P18, P19, P29, P30)

10.1. The arrow of time and emergent expansion (P13)

Where we landed: classical sequential growth gives a clean arrow of time, the number of causal relations accumulates monotonically and irreversibly, the irreversible cascade of the bootstrap. Expansion is demonstrated two ways. Sprinkled into a de Sitter geometry the causal order expands (its intrinsic spatial slices grow with proper time). And, more deeply, expansion EMERGES from a pure local growth rule: with net birth above one (a second child with probability q) the spatial front grows on its own at rate $1 + q$, with a static control at $q = 0$, and no metric imposed. The remaining frontier is a fully emergent spatial geometry, not just an emergent growth rate.

10.2. Inflation (P30)

Where we landed: a time-varying birth rate gives the inflationary profile. A high early rate produces a burst of rapid expansion (4.2 e-folds, the front doubling each generation) followed by a graceful exit to slow expansion (rate about 1.06), all from a local rule with no inflaton field put in by hand.

10.3. Dark energy: the everpresent cosmological constant (P10, P19, P29)

The everpresent model: Λ is conjugate to the spacetime 4-volume V , realized as the element count, which fluctuates as $\sqrt{V}$, so $\delta\Lambda \sim 1/\sqrt{V}$. **Where we landed:** the sharp Benincasa-Dowker action has the fluctuation problem (the implied Λ grows with volume), and smearing tames it. In 2D the smeared action makes the implied Λ shrink with volume (exponent -0.29), the everpresent direction. In 4D the (newly implemented) smeared kernel tames the exponent from $+0.16$ toward zero ($+0.06$), the same direction, though not yet the full shrinking, which likely needs the dynamical conjugate-volume model. The observational payoff is in Section 12.

10.4. Dark matter (P18)

Where we landed: a nonlocal (infrared-enhanced) gravitational kinetic term, natural because the discrete action is nonlocal, flattens galaxy rotation curves with no dark particle. The rotation curve falls under local (Newtonian) gravity (outer-to-inner v^2 ratio 0.23) but stays flat or rises under the nonlocal one (ratio 1.29). This is the mechanism and the flattening, demonstrated, the galactic scale being a separate quantitative question. Two further routes (a gauge-neutral fermion that gravitates but is electromagnetically dark, and relics from the early growth) remain.

11. Holography (P15)

AdS/CFT ties bulk geometry to boundary entropy, and black-hole entropy scales as area, not volume. The microscopic origin is the entanglement across the boundary. **Where we landed:** the free-fermion ground-state entanglement entropy follows the conformal log law in 1D (slope 0.323, central charge $c = 1$) and an area law in 2D (the entropy grows with the boundary, and an area fit beats a volume fit). Entropy living on boundaries rather than volumes is the signature behind holography and black-hole thermodynamics, and it sits naturally with the hyperbolic (AdS-like) substrate of P3. The 3D Bekenstein-Hawking case (entropy by horizon area) is in Section 8 (P33). The full holographic correspondence, a bulk-boundary dictionary and the Ryu-Takayanagi formula, is the long road ahead.

12. Distinctive Predictions and Contact With Data (P26, P27, P35)

A model becomes science when its distinctive predictions meet data. Three of ours do.

12.1. Swerves (P26)

On a continuum geodesic a free particle keeps its velocity forever. On a causal set there is no element exactly straight ahead, so the particle hops to the nearest future element and its momentum gets a tiny random kick each step. **Where we landed:** the rapidity diffuses, its variance growing linearly with proper time (a

diffusion with no continuum analogue, the Dowker-Henson-Sorkin swerve). The diffusion is present at every sprinkling density. The honest caveats: the variance saturates at large proper time (a finite-box effect) and the precise density scaling needs a covariant trajectory rule. The swerve itself is robust.

12.2. Dark energy, the headline match (P35)

The everpresent Λ predicts $\delta\Lambda \sim 1/\sqrt{V}$. At the observed spacetime 4-volume (the past horizon, $(ct/\ell_P)^4$):

- predicted $\delta\Lambda \sim 1.5 \times 10^{-122}$ (Planck units),
- observed $\Lambda \sim 2.9 \times 10^{-122}$ (Planck units),
- ratio 0.53.

Where we landed: the prediction matches the measured dark energy to order of magnitude, within a factor of two. The right value, about 10^{-122} , comes out as one over the square root of the spacetime volume, not as a tuned constant. This is the one quantitative cosmological success of causal-set theory.

12.3. Lorentz violation, a confirmed null (P27, P35)

Gamma-ray-burst photon timing (GRB 090510, Fermi LAT) excludes linear Lorentz violation below about 7.6 times the Planck energy. The framework, using a random sprinkling that is Lorentz-invariant in distribution, predicts *no* linear Lorentz violation at all. **Where we landed:** observation confirms the prediction, and it discriminates, a lattice (or any regular) discreteness would predict a large effect that is already excluded, while the random-sprinkling discreteness predicts the observed null. The discreteness scale is bounded only from below.

12.4. The swerve as a future target

The swerve diffusion is Planck-suppressed, so the accumulated spread over the age of the universe is small, below current bounds (from the narrowness of cosmic-ray and particle momenta). It is a future target for ultra-high-energy cosmic rays rather than a present conflict.

Bottom line. One prediction matches a measured number (dark energy, to order of magnitude), one is confirmed by a null result that excludes the alternatives (no linear Lorentz violation), and one is a clean future target (swerves). This is where the framework makes contact with data, not just internal coherence.

13. The Capstone: The Model Run End-to-End (P34, P36)

Every result above used the substrate best suited to it. The honest question is whether it all falls out of one instantiation of the present model of Section 3.

13.1. The capstone run (P34)

We run that one model, a growing random hyperbolic mesh with the ternary signed-majority rule, and read every key emergent structure off the same mesh. **Where we landed** (one mesh of 1500 vibes):

- the mesh has mean degree 10.6, Lorentz anisotropy 0.070 (no preferred frame), and exponential reach,

- the dynamics keeps tones strictly ternary and converges to a stable structured state (the flip fraction falls to zero), a non-trivial fixed point in the manner of a Hopfield network, the basis for persistent matter and selves,
- the emergent Hamiltonian (the graph Laplacian) is bounded below (minimum eigenvalue $0.007 \geq 0$) and local (range 1),
- the arrow of time holds, relations only accumulate as the mesh grows.

So the model runs as one system. From a single instantiation come a definite geometry with no preferred frame, stable structured states, the local bounded-below Hamiltonian, and the arrow. The matter, force, and gravity sectors are read off the same mesh by their operators. The natural next integration is to evolve all of them by the one microscopic rule in a single simulation.

13.2. The model written at a glance (P36)

The present model is written and run by a small constructor, so its description and its implementation are one object. With no options it is the present model, and a one-word change expresses a variant for comparison:

```
const model = vibe().size(1500).seed(1)    // the present model
console.log(model.describe())              // print it at a glance
const world = model.build().run(40)        // build the mesh, run 40 beats
world.read()                              // emergent structures off the mesh
```

Swapping `vibe().mesh('lattice')` raises the Lorentz anisotropy from 0.06 to 1.0 (a strong preferred frame), making plain why the random hyperbolic mesh is the one chosen: the lattice variant breaks Lorentz invariance and the hyperbolic one does not. This is the model definition the rest of the paper rests on, legible and executable at once.

14. The Full Scoreboard

Every finding, with its honest status. The status words: *validated* (a stated prediction confirmed by the testbed), *measured* (a previously bounded quantity now computed), *demonstrated* (the mechanism and signature shown), *down-payment* (a first rung on a deep, still-open problem), *progress* or *clarified* (partial or corrected), *boundary* (outside what a simulation can decide).

ID	Question	Where we landed
P1	local bounded-below Hamiltonian	resolved: a trilemma, solved by the emergent-mesh Laplacian (local, bounded below, propagating)
P2	a dynamics favoring manifold order	solved at scale: a first-order transition, smeared action makes spacetime a stable phase
P3	addressing versus Lorentz	candidate solved: the hyperbolic random mesh is reaching, Lorentz-safe, and navigable at once
P4	the monist spinor and chirality	validated: zero modes equal the Betti sum, the overlap threads the chirality wall

continued on the next page

ID	Question	Where we landed
P5	the Hauptvermutung	validated: geometry recovered sharply (3.02 ± 0.05), proof open
P6	a computable 2D path integral	solved at scale: the stable manifold phase is genuinely 2D
P7	quantum from a classical base	quantified: aligned bits not bits, with the separation-decay residual
P8	a gauge field and a charged fermion	validated: confinement, the index theorem, the chiral condensate
P9	structure and experience	boundary: only the structural correlates are measurable
P10	the cosmological constant	progress: the smeared action shrinks the implied Λ with volume
P11	Lorentz invariance of the dynamics	clarified: rotational invariance emerges in the infrared on both substrates
P12	the free-energy crossing	measured: $\beta^* \approx 0.14$, the manifold phase dominates above it
P13	the arrow of time and cosmology	validated: the arrow holds, expansion from geometry and from a growth rule
P14	mass and the spectral gap	validated: gap equals the mass, dispersion $E^2 = p^2 + m^2$
P15	the entanglement area law	validated: the 1D conformal log law and the 2D area law
P16	the Newtonian limit	validated: the static potential is $1/r$ in 3D ($R^2 = 0.997$)
P17	quantum coherence	validated: a quantum walk is ballistic, a classical one diffusive
P18	dark matter	mechanism shown: nonlocal gravity flattens the rotation curve, no dark particle
P19	dark energy in 4D	measured: the sharp action's fluctuation problem, tamed by smearing (P29)
P20	the photon	validated: the free U(1) field is massless, gauge-invariant, two polarizations
P21	the graviton	validated: massless spin-2, two transverse-traceless polarizations
P22	the Higgs	validated: symmetry breaking gives a photon mass $(gv)^2$
P23	the gauge operator from the action	validated: the Maxwell operator is the small-field limit of the Wilson action
P24	the graviton from the action	validated: the linearized Einstein operator, diffeo-invariant, two massless modes
P25	electroweak breaking	validated: $SU(2) \times U(1) \rightarrow U(1)_{\text{EM}}$, the observed W and Z masses
P26	swerves	demonstrated: momentum diffusion from discreteness, no continuum analogue
P27	Lorentz violation	distinctive: the random mesh is Lorentz-safe, predicting the observed null
P28	singularity resolution	demonstrated: discreteness caps the curvature, no infinities
P29	dark energy with the smeared kernel	progress: the 4D smeared kernel tames the fluctuation toward shrinking

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ID	Question	Where we landed
P30	inflation	demonstrated: a time-varying birth rate gives a burst with a graceful exit
P31	the quantum formalism	down-payment: unitarity, interference, and a conserved Born rule
P32	the Einstein equations	down-payment: transverse G (conservation), Newton, a c -speed graviton
P33	black-hole entropy	demonstrated: entropy scales with horizon area, Bekenstein-Hawking
P34	the capstone, the model run end-to-end	demonstrated: one mesh and one rule yield all the key structures
P35	contact with data	meets observation: the dark-energy match, the confirmed Lorentz null
P36	the model DSL	a tool: the present model in a few fluent lines, building and running

14.1. What is and is not established

Demonstrated and validated: the geometry and its dominance, the law, the Lorentz-safe substrate, the field content with mass, the gauge and graviton operators from the action, the electroweak masses, the Newtonian limit, the conserving Einstein equation, singularity resolution, the area law, the quantum pillars, the arrow, expansion, inflation, and that the one model runs end-to-end.

Meets observation: the everpresent Λ matches the dark-energy magnitude to order of magnitude, and the absence of linear Lorentz violation is confirmed.

Open or beyond reach: the Born-rule derivation, the full nonlinear Einstein equation, the chiral gauge theory and the rest of the Standard Model, Hawking radiation, the large- N free-energy crossing, a fully emergent spatial geometry from a growth rule, and the framework’s foundational claim about experience.

15. Conclusion

This paper has given a technical account of Vibe Theory as it now stands: the model in its current form, and thirty-six findings from a finite, reproducible testbed.

The model is one legible object. It treats reality as a single kind of entity, the vibe, a unit of experience carrying a ternary tone and related only by the note, on a growing, random, hyperbolic, causal mesh, updated by a local ternary signed-majority rule. There are no continuous weights and no hidden state, and every continuous quantity of physics is an emergent, coarse-grained description of the discrete substrate. The model is written and run at a glance by a small constructor, so its description and its implementation are the same object.

In one picture, the structure is a growing, random, hyperbolic network of three-valued points that vote locally, a causal set carrying a network automaton. It is deliberately not a regular grid or a regular tiling, which would single out a preferred frame and break Lorentz invariance, and not a field on a smooth space, since

there is none underneath. The smooth quantities of physics, the geometry, the energy and flow of time, and the quantum amplitude, are large-scale summaries of the web rather than values stored on any point. That single object is what every finding above was measured on.

The findings, taken together, make two claims. Monism is computationally coherent and expressive: geometry, time, the law, matter and the four field spins, mass, force, gravity, the quantum pillars, the dark sector, and cosmology all come out of the one mesh, with the gauge and graviton operators derived from the action and the electroweak masses reproduced. And the model runs as one system: a single instantiation of the present model yields a definite Lorentz-safe geometry, stable structured states, the local bounded-below Hamiltonian, and the arrow of time. That is the difference between a pile of demonstrations on convenient substrates and one model that works end-to-end.

Three of the framework’s distinctive predictions make contact with data. The everpresent cosmological constant matches the observed dark energy to order of magnitude, the right value emerging as one over the square root of the spacetime volume rather than a tuned constant. The absence of linear Lorentz violation is confirmed by gamma-ray-burst timing, and it follows from the random sprinkling in a way a lattice could not survive. And the swerve is a clean future target.

We have been equally precise about the limits. The Born rule is shown consistent, not derived. The Einstein equation is linearized, not yet nonlinear. The Standard-Model content beyond the electroweak masses, Hawking radiation, the large- N free-energy crossing, and a fully emergent spatial geometry from a growth rule are open. The framework’s foundational claim about experience is a posit, outside what any simulation can decide.

What Vibe Theory offers, then, is a candidate ontology that has been written down as one legible model, simulated rather than only argued, validated against known physics where it can be, and confronted with data where its predictions are distinctive. Its physical ladder is built and largely validated, its hardest dynamical question is measured, its quantum question is sharpened to a precise open problem, two of its predictions already meet observation, and its boundary is named honestly. The substrate is one kind of thing, related to itself in different patterns, and a great deal of the world really does come out of it.

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